



Lec-104 Effect on Bandwidth of Multistage Amplifier in Int...

R K Classes

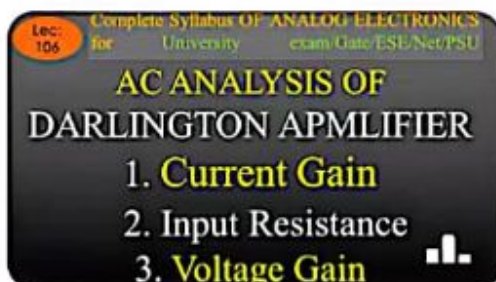
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Lec-105 Introduction to Darlington Pair Amplifier wit...

R K Classes

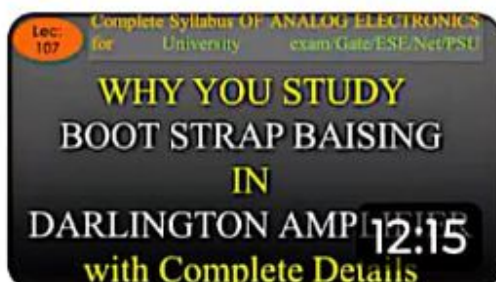
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Lec-106 AC Analysis of Darli...

R K Classes

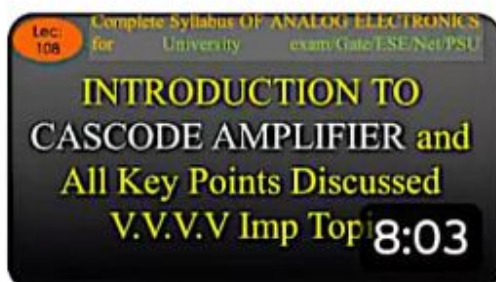
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Lec-107 Why Use Boot Strap Biasing in Darlington Pair A...

R K Classes

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Lec-108 Introduction to Cascode Amplifier with Cha...

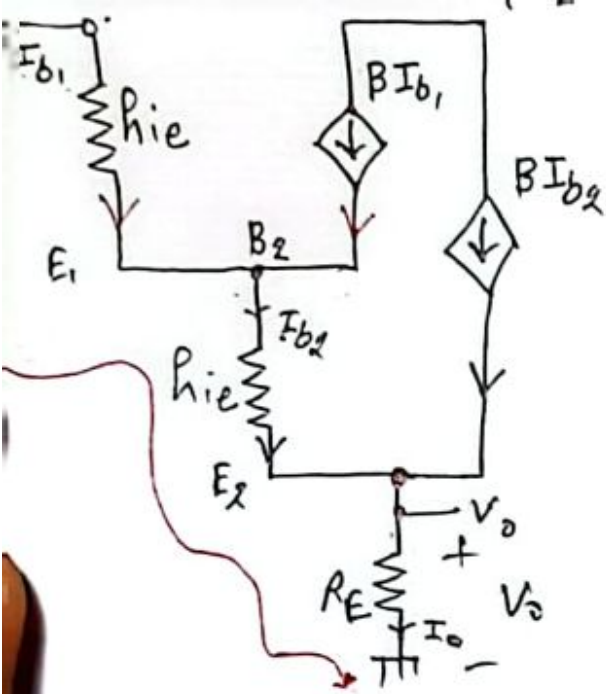
R K Classes

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Lec-109 Solved Problem





① Input Impedance Z_{in}

$$V_{in} = I_{b1} h_{ie} + I_{b2} h_{ie} + I_o R_E$$

$$V_{in} = I_{b1} h_{ie} + I_{b1} (1+\beta) h_{ie} + I_{b1} (1+\beta)^2 R_E$$

$$\frac{V_{in}}{I_{b1}} = h_{ie} + (1+\beta) h_{ie} + (1+\beta)^2 R_E$$

$$R_{in} \approx (1+\beta)^2 R_E$$

current gain

$$A_I = \frac{I_o}{I_{in}} = \frac{I_o}{I_{b1}}$$

$$I_o = I_{b2} + \beta I_{b2} = I_{b2} (1+\beta)$$

$$I_{b2} = I_{b1} + \beta I_{b1}$$

$$I_{b2} = I_{b1} (1+\beta)$$

$$I_o = I_{b1} (1+\beta) (1+\beta)$$

$$\frac{I_o}{I_{b1}} = (1+\beta) (1+\beta) = (1+\beta)^2$$

$$A_I \approx \beta^2$$

③ Voltage Gain $A_v = \frac{V_o}{V_{in}}$

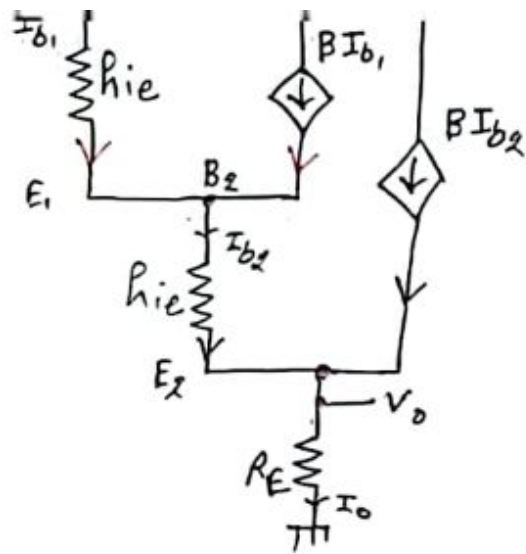
$$V_o = I_o \times R_E$$

$$V_o = I_{b1} \times (1+\beta)^2 R_E$$

$$V_{in} = I_{b1} h_{ie} + I_{b1} (1+\beta) h_{ie} + I_{b1} (1+\beta)^2 R_E$$

$$\frac{V_o}{V_{in}} = \frac{I_{b1} (1+\beta)^2 R_E}{I_{b1} [h_{ie} + (1+\beta) h_{ie} + (1+\beta)^2 R_E]}$$

$$\frac{V_o}{V_{in}} = \frac{(1+\beta)^2 R_E}{h_{ie} + (1+\beta) h_{ie} + (1+\beta)^2 R_E}$$



③ Voltage Gain A_v

① current gain

$$A_I = \frac{I_o}{I_{in}} = \frac{I_o}{I_{b1}}$$

$$I_o = I_{b2} + \beta I_{b2} = \underline{I_{b2}(1+\beta)}$$

$$I_{b2} = I_{b1} + \beta I_{b1}$$

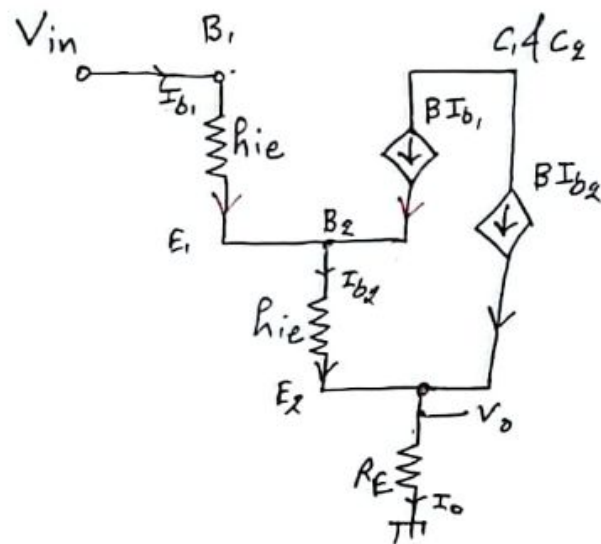
$$\underline{I_{b2} = I_{b1}(1+\beta)}$$

$$I_o = I_{b1}(1+\beta)(1+\beta)$$

$$\boxed{\frac{I_o}{I_{b1}} = (1+\beta)(1+\beta)} \stackrel{\text{KVA}}{=} (1+\beta)^2$$

$$A_I \approx \beta^2$$

AC Analysis of Darlington Amplifier



① Input Resistance (z_{in})

③ Voltage Gain A_v

① current gain

$$A_I = \frac{I_o}{I_{in}} = \frac{I_o}{I_{b1}}$$

$$I_o = I_{b2} + \beta I_{b2} = \underline{I_{b2}(1+\beta)}$$



Lec-103 Effect on Bandwidth of Multistage Amplifier in N...



R K Classes

4.9K views · 3 years ago



Lec-104 Effect on Bandwidth of Multistage Amplifier in Int...



R K Classes

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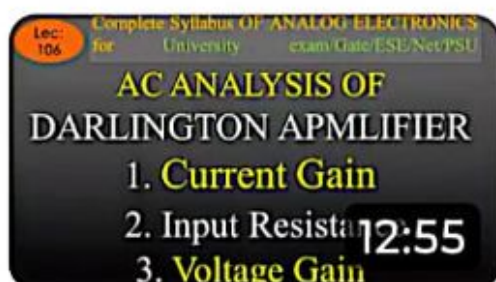


Lec-105 Introduction to Darli...



R K Classes

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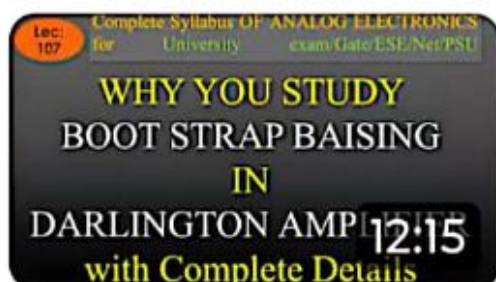


Lec-106 AC Analysis of Darlington Pair Amplifier usi...



R K Classes

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Lec-107 Why Use Boot Strap Biasing in Darlington Pair A...



R K Classes

8.8K views · 3 years ago



Lec-108 Introduction to



Lec-109 Solved Problem Based on Multistage Amplifier Part-1 | A E | R K ...

5.3K views 3 yr ago #RKClasses ...more



R K Classes 25.2K

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Lec:
110

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for University exam/Gate/ESE/Net/PSU

SOLVED PROBLEM ON MULTISTAGE AMPLIFIER PART-2

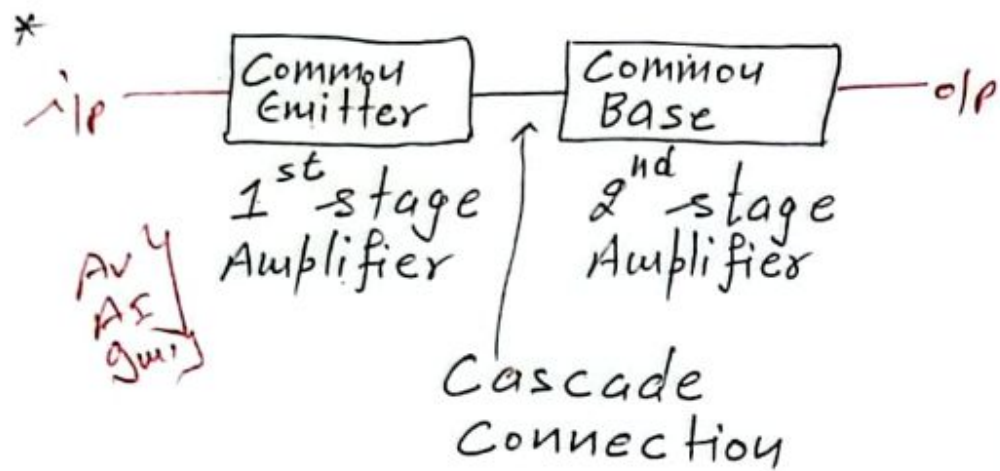
6:51



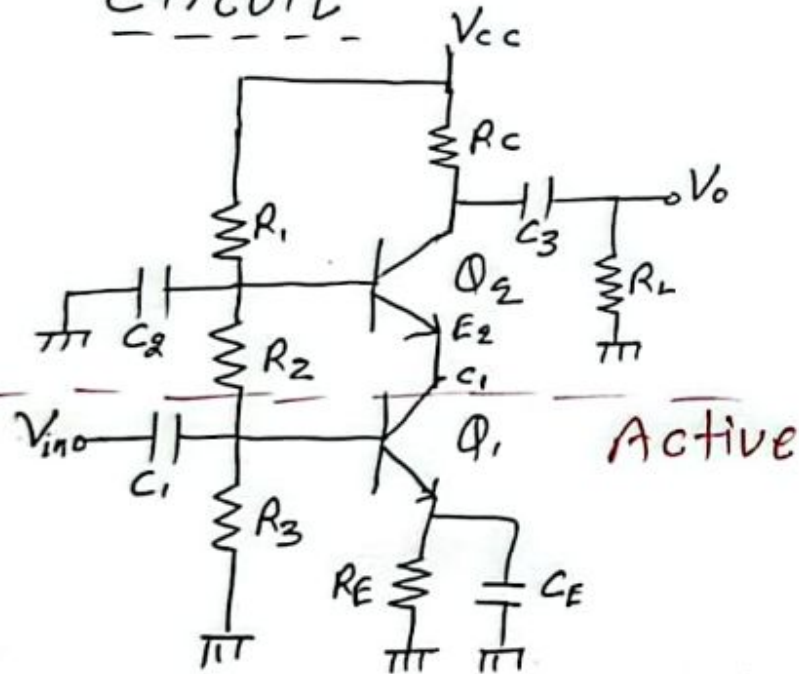
Lec-110 Solved Problem Based on
Multistage Amplifier Part-2 | A E | R K Clas...



R K Classes · 2.6K views · 3 years ago



Circuit



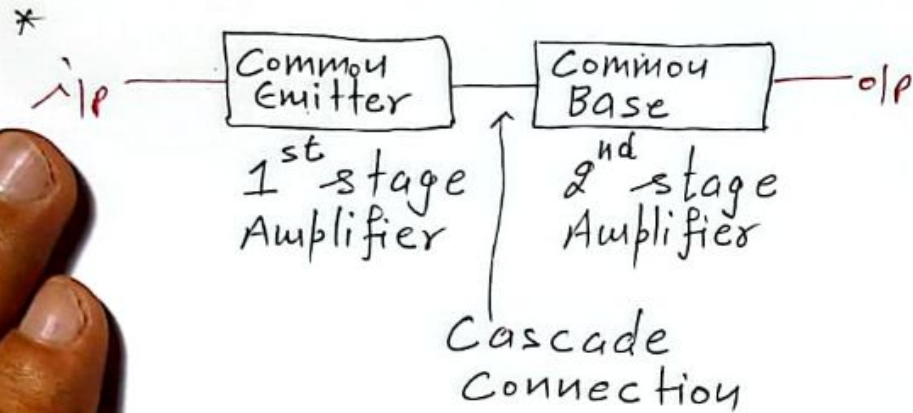
$Q_1 \rightarrow$ CE configuration
 $Q_2 \rightarrow$ CB "

- ① It has large bandwidth (BW)
- ② Wide band Amplifier
- ③ known as CE-CB Amplifier BJT
or CS-CG Amplifier FET
- ④ It is used as Video freq^y Amplifier
- ⑤ Input Resistance is small = h_{ie}
- ⑥ $A_I = -\beta$ → Same as CE configuration
- ⑦ $A_V = -\frac{\beta R_L'}{h_{ie}}$ → Same as " " "
- ⑧ overall transconductance of Cascode Amplifier = g_{m1}

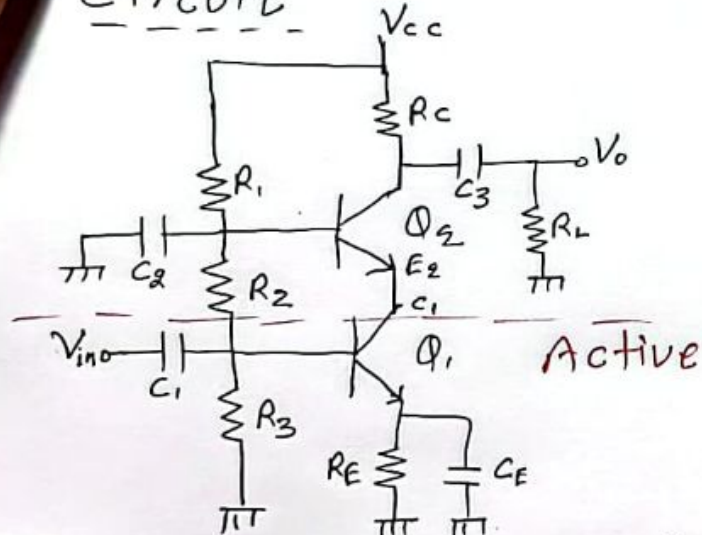
Homework:- AC Analysis of Cascode

Cascode Amplifier

* CE & CB Amplifier



Circuit



Q. → CE Configuration

Key Points

- ① It has large bandwidth (BW)
- ② Wide band Amplifier
- ③ known as CE-CB Amplifier BJT or CS-CG Amplifier FET
- ④ It is used as Video freq^y Amplifier
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- ⑧ overall transconductance of Cascode Amplifier =

Homework:- AC Analysis

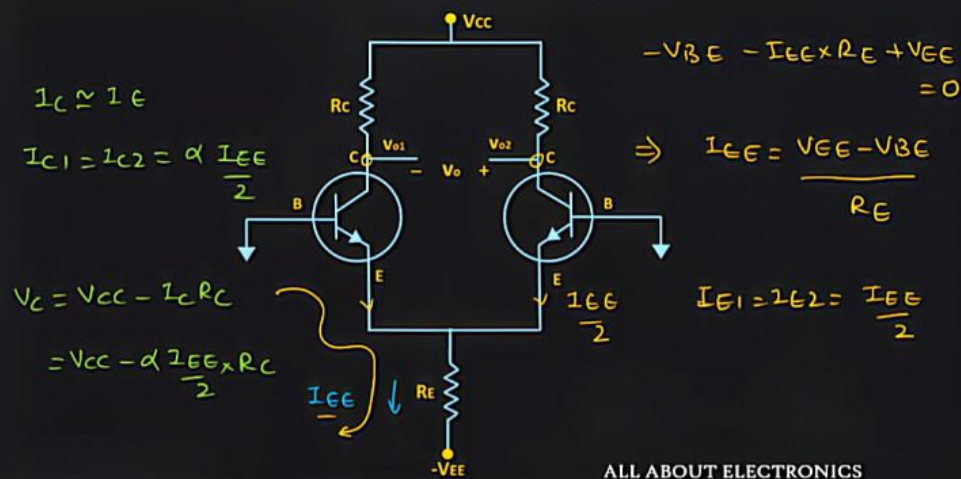
MOSFET Amplifier with Active Load



$$\text{Intrinsic Gain } |A_o| = g_m r_o$$

So we have seen that, typically in the ICs, the achievable intrinsic gain is in the range

BJT Differential Amplifier



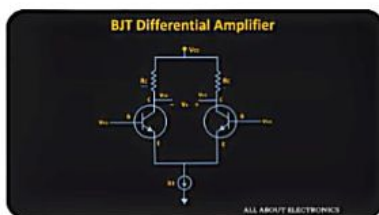
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Amplifier

0:51



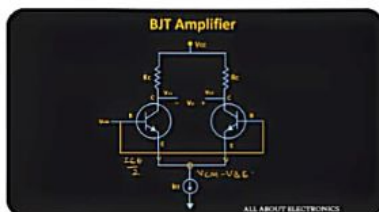
BJT- Differential Amplifier circuit

3:00



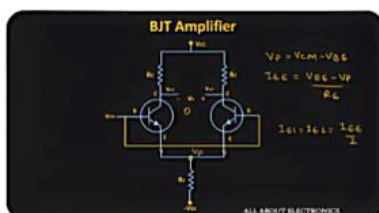
DC analysis of Differential Amplifier

4:18



Response of Differential Amplifier to Common Mode input signal (Qualit...

6:21

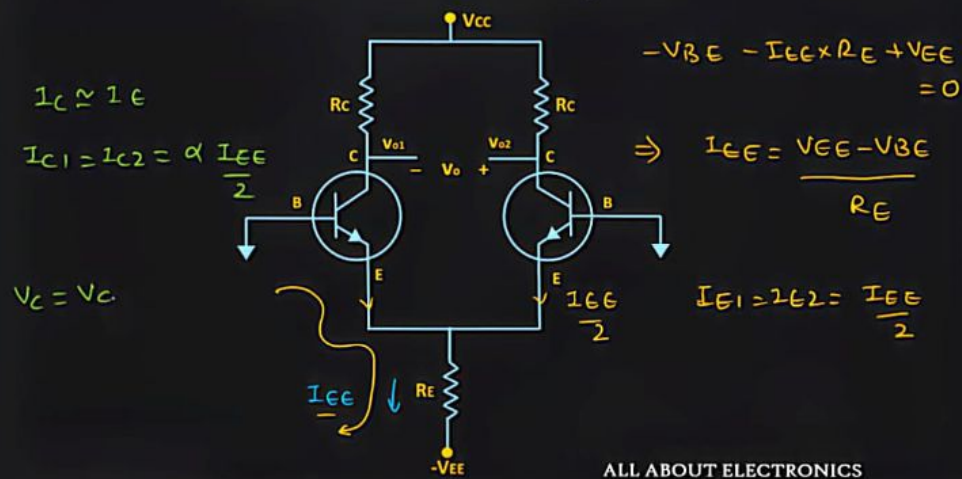


Response of Differential Amplifier to Differential Input signal (Qualitative ...

9:00

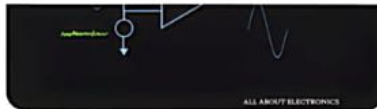
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BJT Differential Amplifier



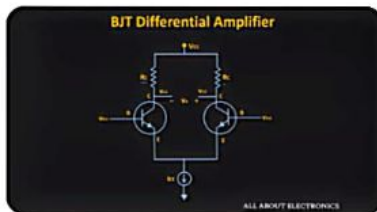
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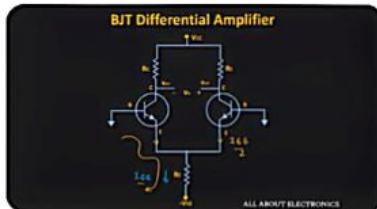
Amplifier

0:51



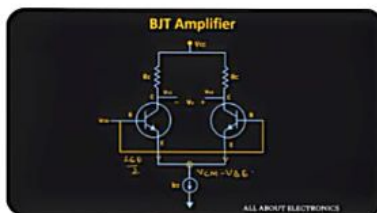
BJT- Differential Amplifier circuit

3:00



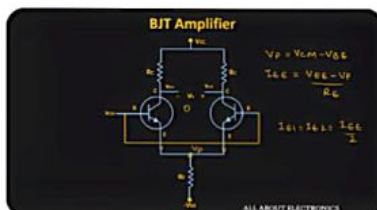
DC analysis of Differential Amplifier

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Response of Differential Amplifier to Common Mode input signal (Qualit...

6:21

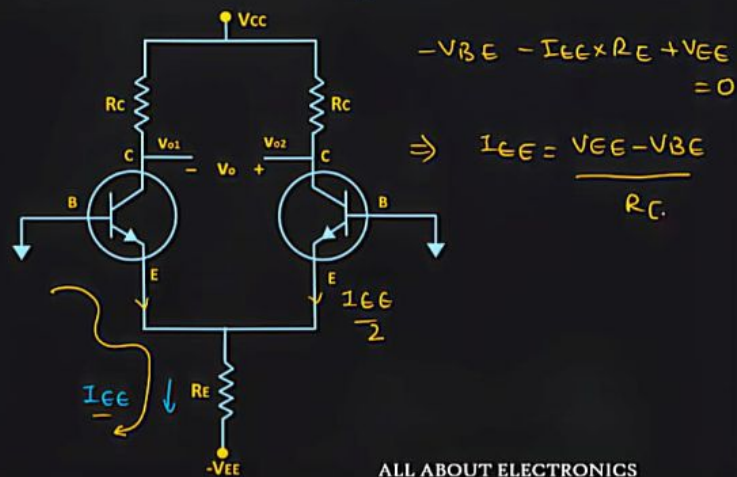


Response of Differential Amplifier to Differential Input signal (Qualitative ...

9:00

Sync to video time

BJT Differential Amplifier



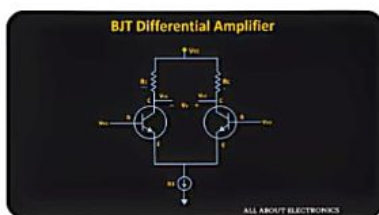
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Amplifier

0:51



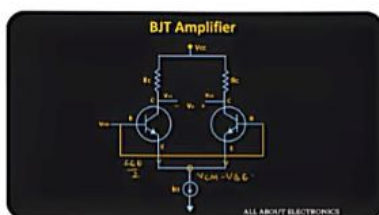
BJT- Differential Amplifier circuit

3:00



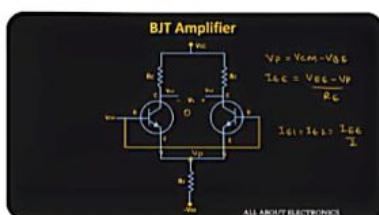
DC analysis of Differential Amplifier

4:18



Response of Differential Amplifier to Common Mode input signal (Qualit...

6:21

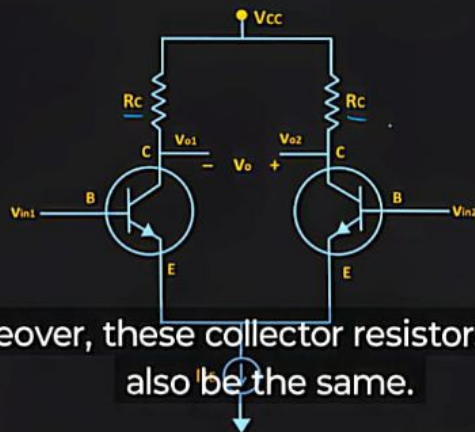


Response of Differential Amplifier to Differential Input signal (Qualitative ...)

9:00

Sync to video time

BJT Differential Amplifier



Moreover, these collector resistors should also be the same.

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BJT Differential Amplifier

Introduction

0:00

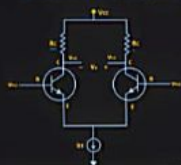
Differential Amplifier



Advantages of Differential Amplifier

0:51

BJT Differential Amplifier



BJT- Differential Amplifier circuit

3:00



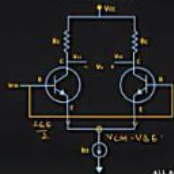
BJT Differential Amplifier



DC analysis of Differential Amplifier

4:18

BJT Amplifier



Response of Differential Amplifier to Common Mode input signal (Qualit...

Sync to video time



Question-01

Calculate the CE amplifiers with following specification $\beta=100$, $g_m = 386 \text{ A/V}$, $r_o = \infty$, $r_\pi = 259 \Omega$, $r_s = 1 \text{ k}\Omega$, $R_B = 93 \text{ k}\Omega$, $R_C = 250 \Omega$, $R_L = 1 \text{ k}\Omega$, $R_1 = \infty$, $C_2 = 4.7 \mu\text{F}$.

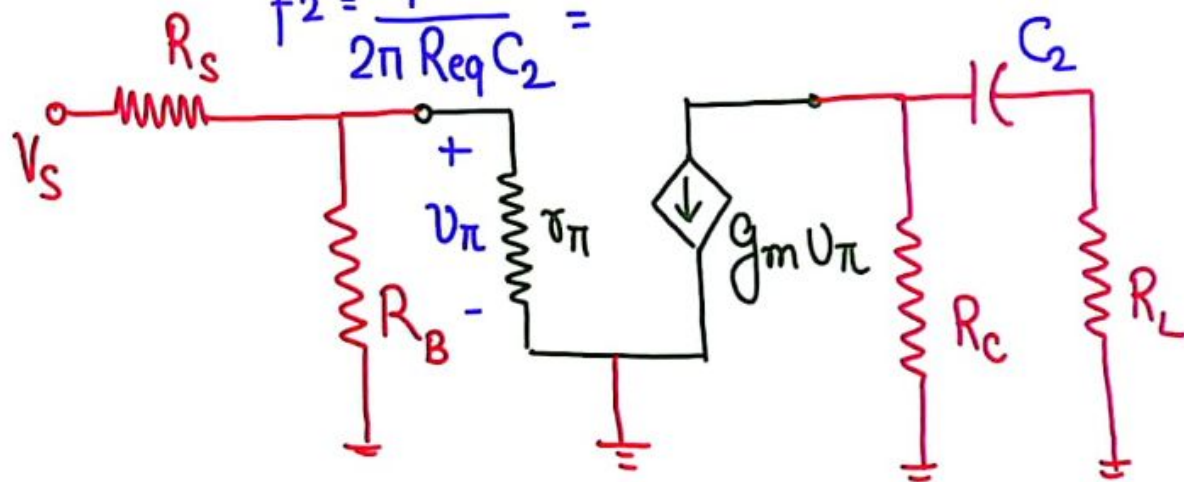
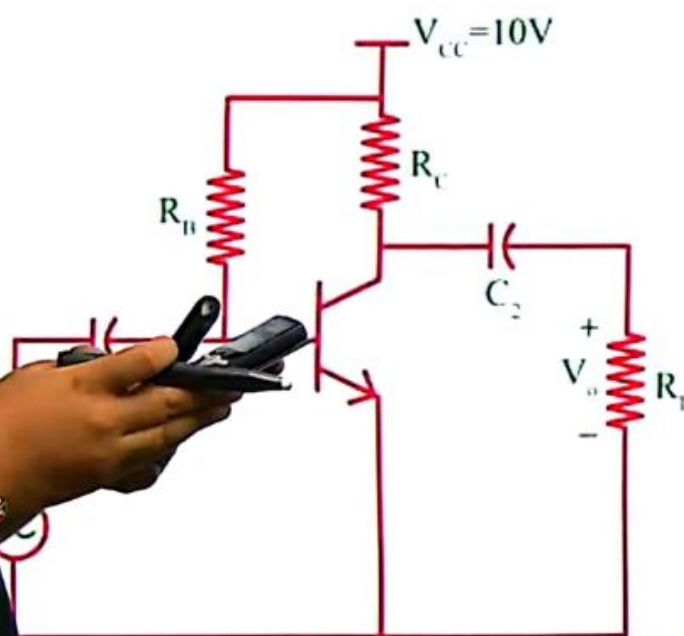
(1) Find the resistance seen by source V_S .

(2) Find the lower cut-off frequency due to C_2 .

$$R_S + R_B \parallel r_\pi = 1.258 \text{ k}\Omega$$

$$R_{eq} = R_C + R_L = 1.25 \text{ k}\Omega$$

$$f_2 = \frac{1}{2\pi R_{eq} C_2} =$$

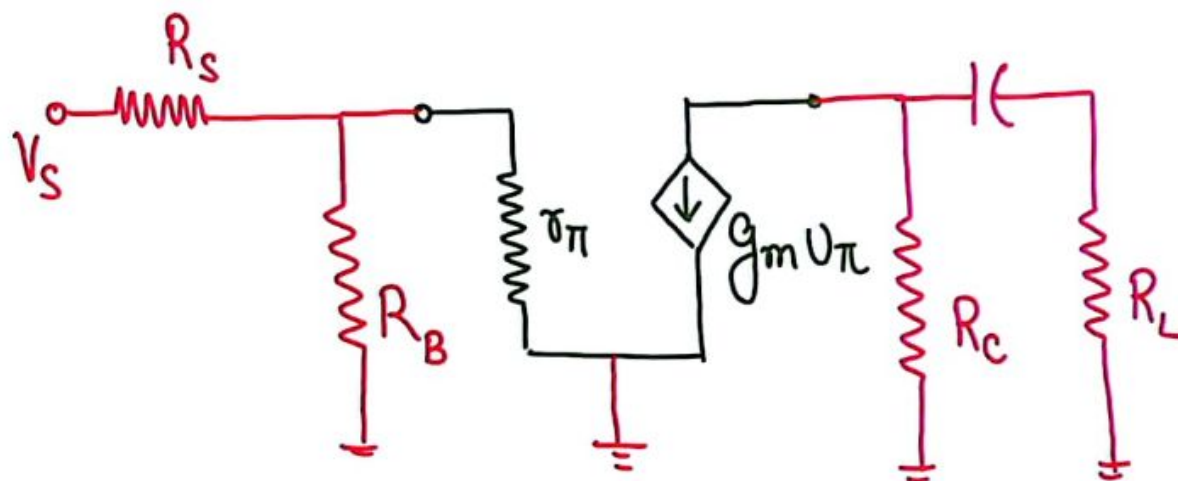
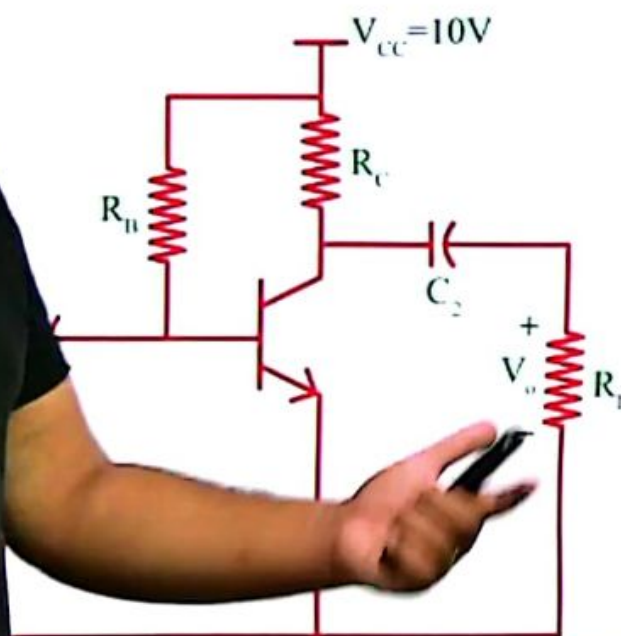




Question-01

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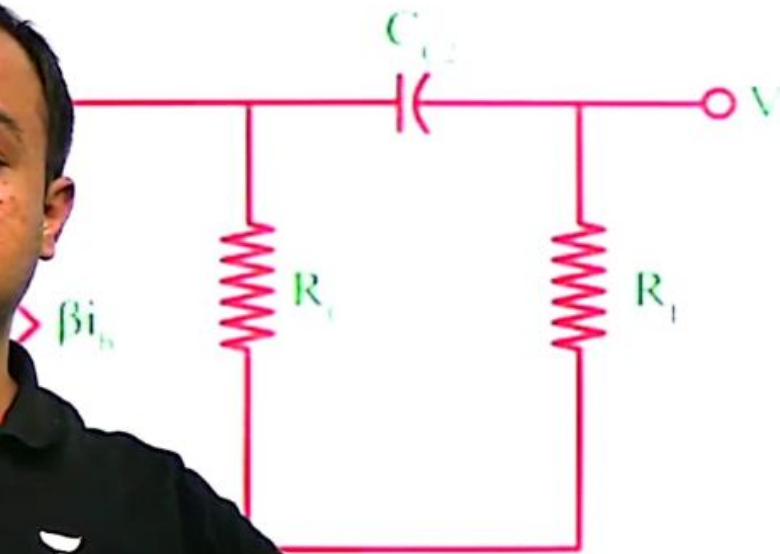
- (1) Find the resistance seen by source V_S .
- (2) Find the lower cut-off frequency due to C_2 .





Effect of C_{C2}

Output side circuit:



short C_{C1} , C_E

ground V_s , $i_b = 0$ $\beta i_b = 0$

resistance across $C_{C2} = R_C + R_L$

$$f_{L3} = \frac{1}{2\pi R_3 C_{C2}}$$

$$f_L = \max [f_{L1}, f_{L2}, f_{L3}]$$

$$V_o = - \frac{\beta v_{\pi} (\tau_o \parallel R_c)}{\tau_{\pi}}$$

$$V_o = - \frac{\beta (\tau_o \parallel R_c)}{\tau_{\pi}} \left[\frac{R_1 \parallel R_2 \parallel \tau_{\pi}}{R_1 \parallel R_2 \parallel \tau_{\pi} + R_s + 1/sC_{c1}} \right]$$

$$= - \frac{\beta (\tau_o \parallel R_c)}{\tau_{\pi}} \frac{[sC_{c1} (R_B \parallel \tau_{\pi})]}{1 + sC_{c1} [R_s + R_B \parallel \tau_{\pi}]}$$

$$R_B = R_1 \parallel R_2$$

$$= - \frac{\beta (\tau_o \parallel R_c)}{\tau_{\pi}} \times \frac{R_B \parallel \tau_{\pi}}{R_s + R_B \parallel \tau_{\pi}} \left[\frac{sC_{c1} [R_s + R_B \parallel \tau_{\pi}]}{1 + sC_{c1} [R_s + R_B \parallel \tau_{\pi}]} \right]$$

Cut off frequency

$$\omega_L = \frac{1}{(R_s + R_B \parallel \tau_{\pi}) C_{c1}}$$

$$\text{gain} = A_v(\text{mid}) \times \frac{sCR}{1 + sCR}$$

Similar to

HIPF

$$V_o = - \frac{\beta v_{\pi} (\tau_o \parallel R_c)}{\tau_{\pi}}$$

$$\frac{V_o}{V_s} = - \frac{\beta (\tau_o \parallel R_c)}{\tau_{\pi}} \left[\frac{R_1 \parallel R_2 \parallel \tau_{\pi}}{R_1 \parallel R_2 \parallel \tau_{\pi} + R_s + 1/sC_{c1}} \right]$$

$$= - \frac{\beta (\tau_o \parallel R_c)}{\tau_{\pi}} \frac{[sC_{c1} (R_B \parallel \tau_{\pi})]}{1 + sC_{c1} [R_s + R_B \parallel \tau_{\pi}]}$$

$$R_B = R_1 \parallel R_2$$

Cut off frequency

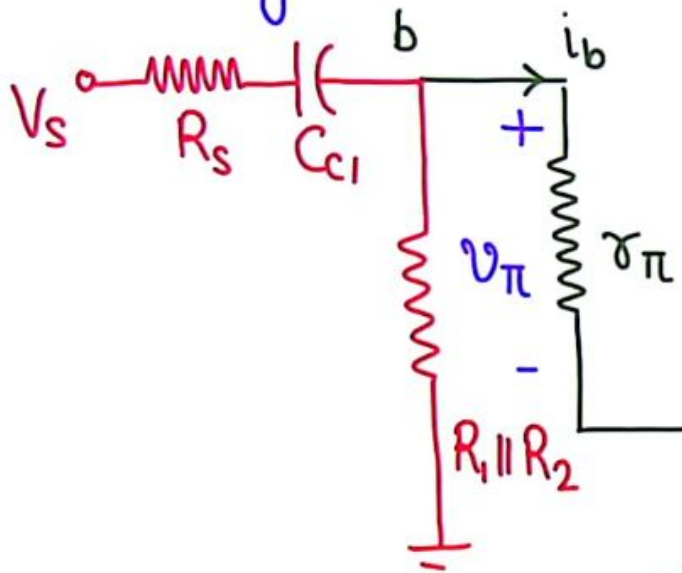
Similar to
HPF





Effect of C_{c1}

using r_e model



$$V_o = -\beta i_b (r_o \parallel R_c)$$

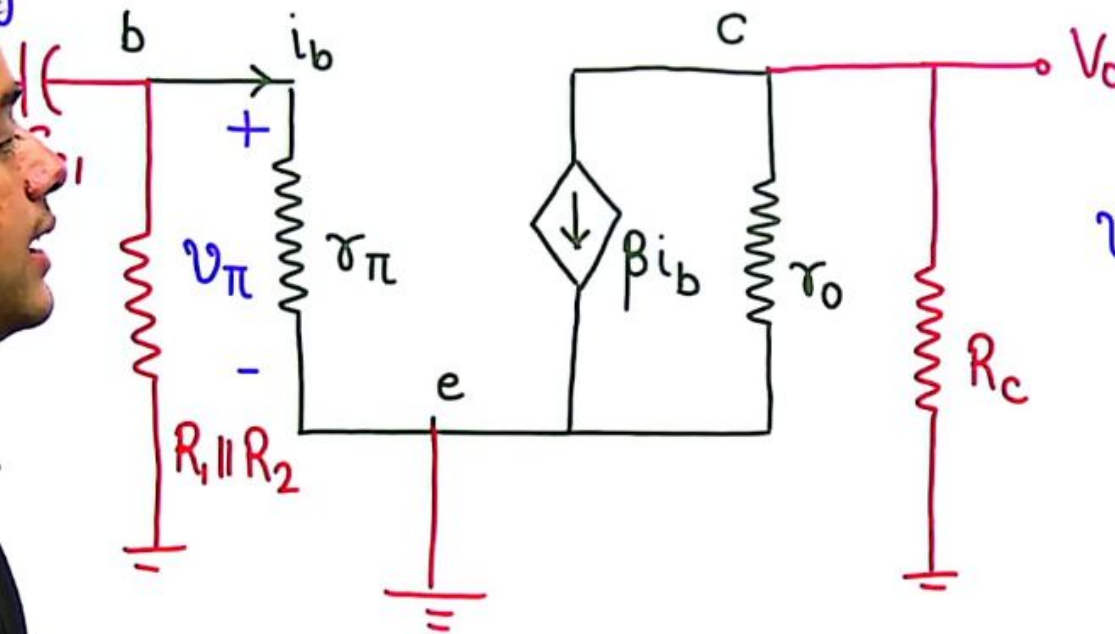
$$v_{\pi} = \frac{V_s [R_1 \parallel R_2 \parallel r_{\pi}]}{[R_1 \parallel R_2 \parallel r_{\pi} + R_s + \frac{1}{sC_{c1}}]}$$

$$= v_{\pi} / r_{\pi}$$



Effect of C_{C1}

using r_e model



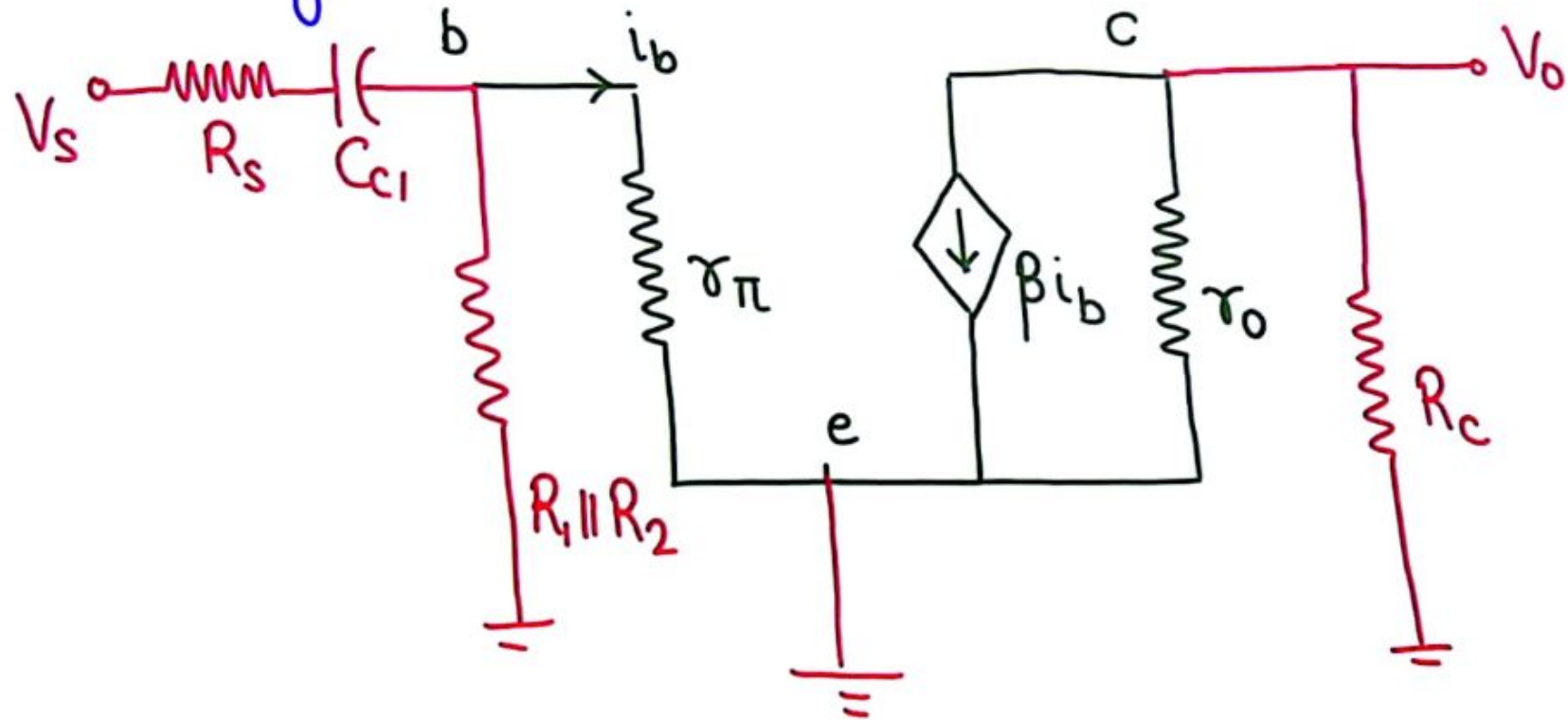
$$V_o = -\beta i_b (r_o \parallel R_c)$$

$$v_{\pi} = \frac{V_s [R_1 \parallel R_2 \parallel r_{\pi}]}{[R_1 \parallel R_2 \parallel r_{\pi} + R_s + \frac{1}{sC_{C1}}]}$$



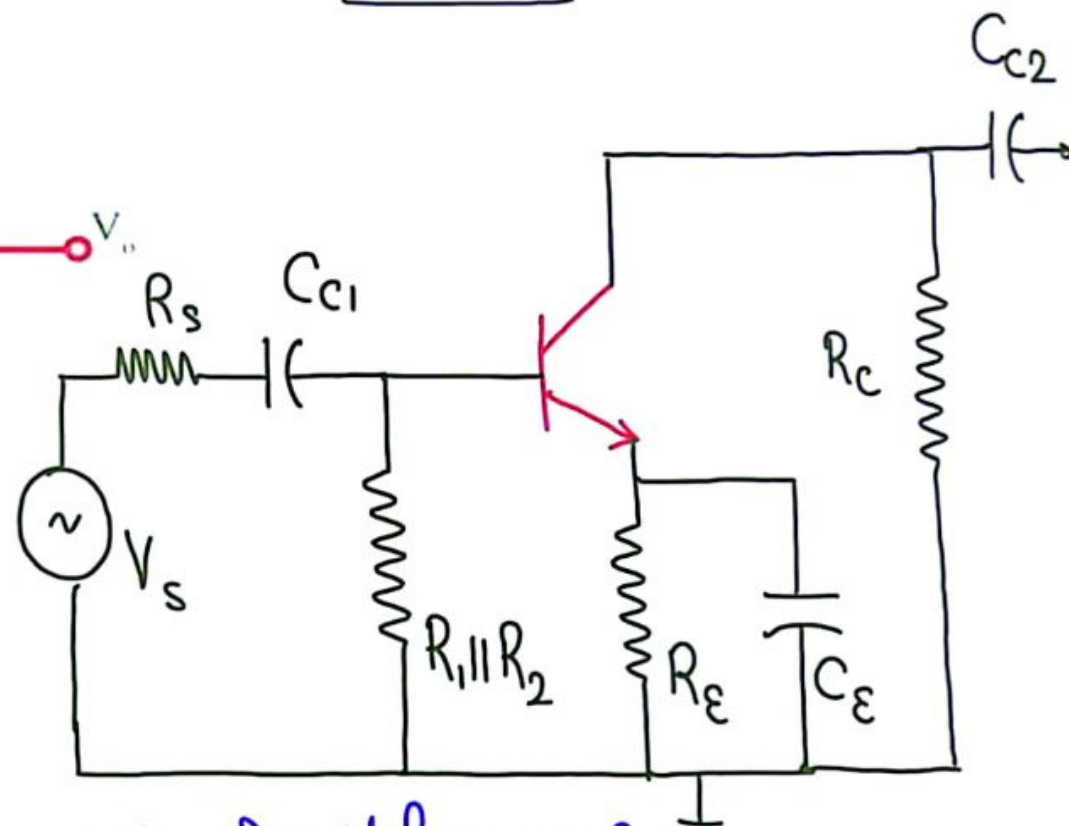
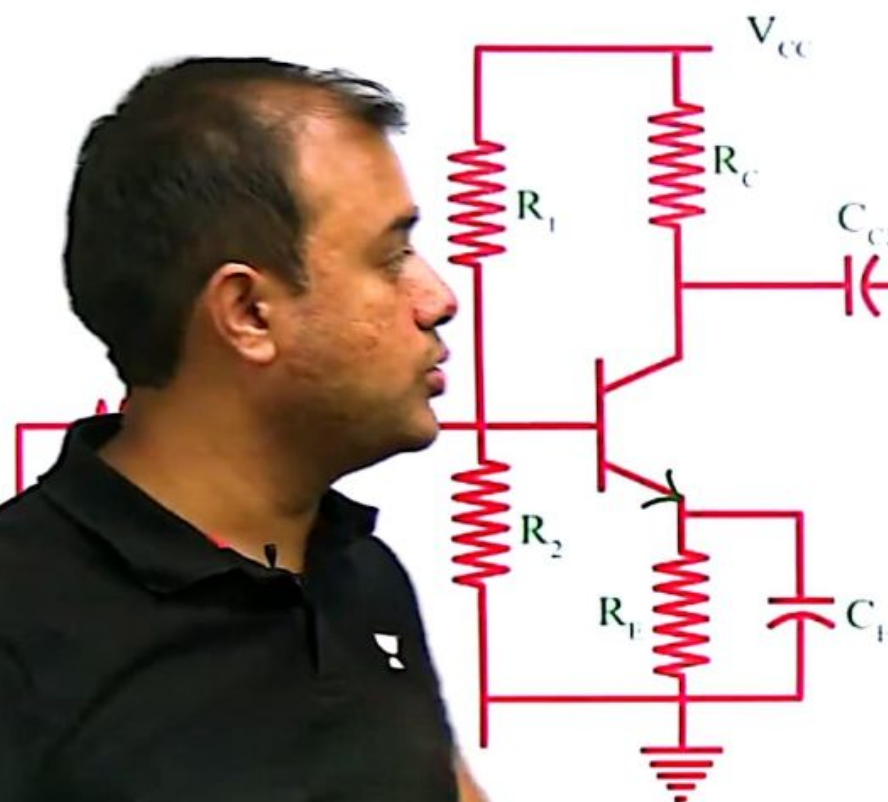
Effect of C_{C1}

using r_e model



(1) Low Frequency Response Of Common Emitter Amplifier

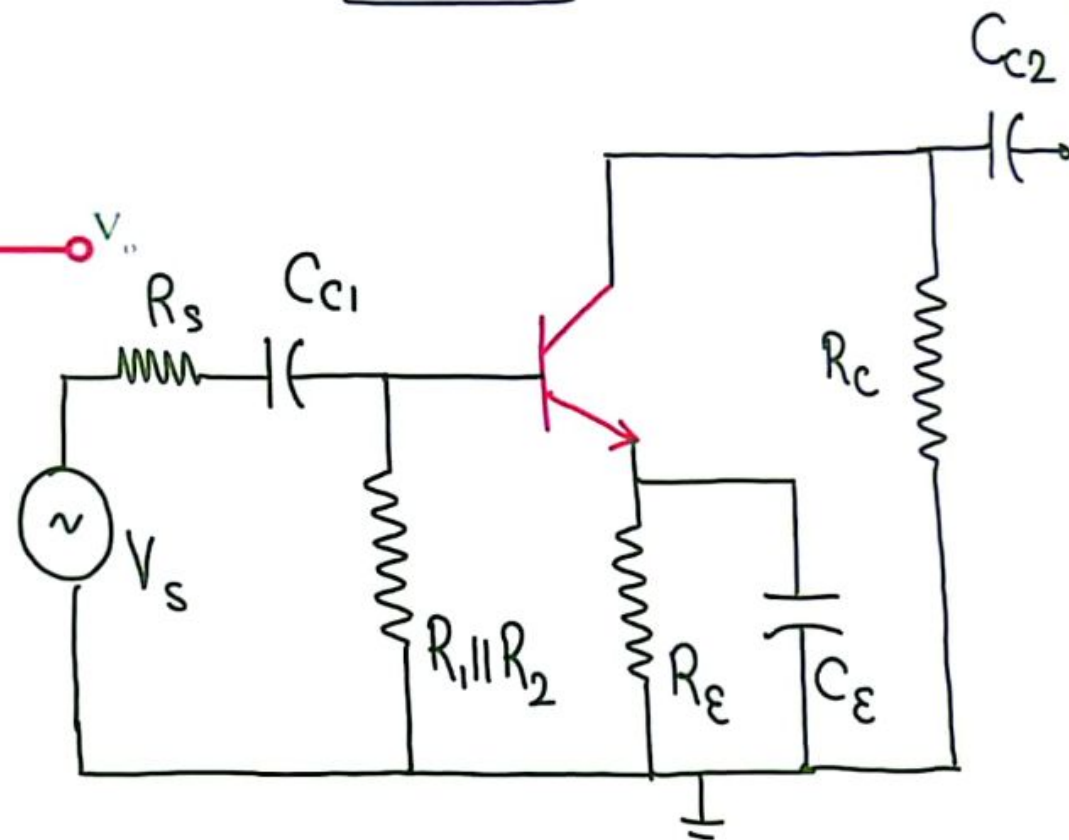
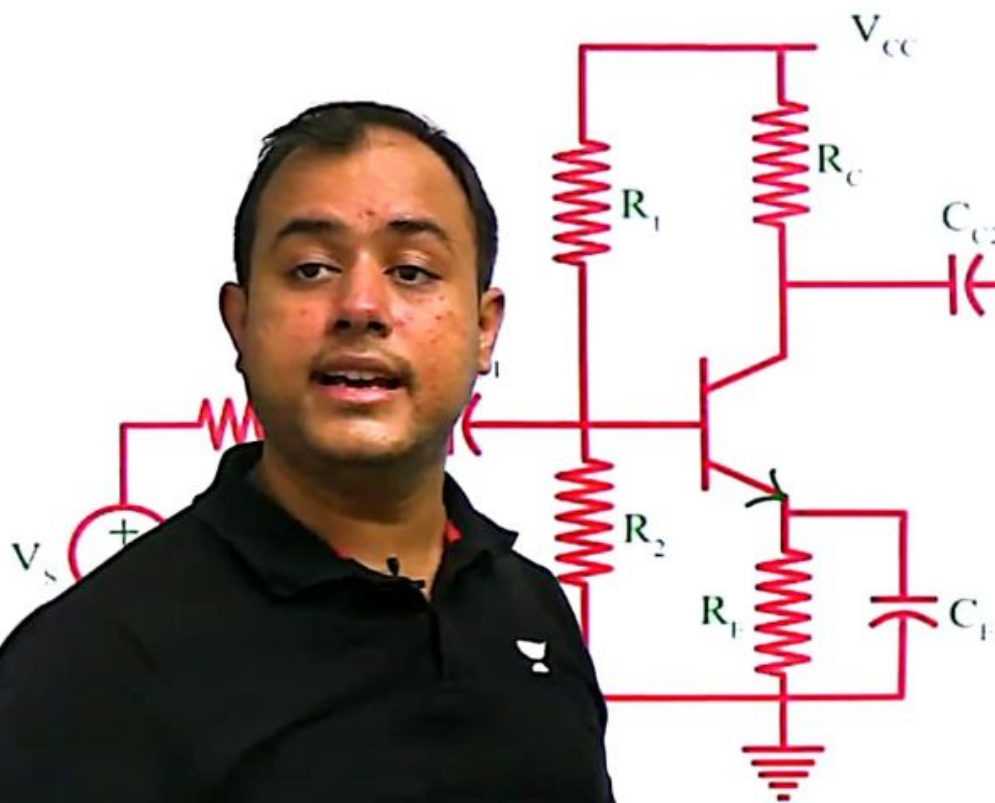
ac model



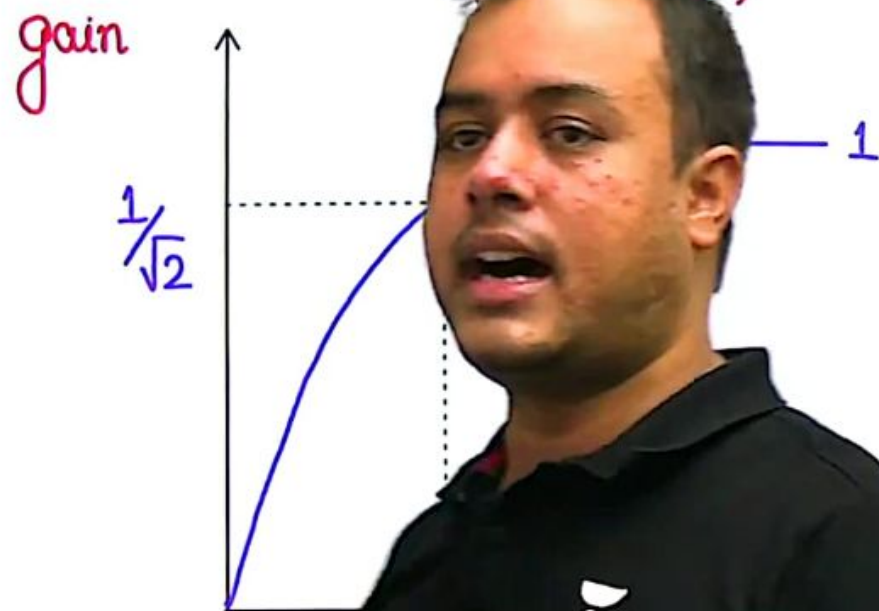
in low & mid freq we can
use any model

(1) Low Frequency Response Of Common Emitter Amplifier

ac model



$$\text{Gain (HPF)} = \frac{1}{\sqrt{1 + (f_L/f)^2}}$$



$$f_L = \frac{1}{2\pi RC}$$

R = equivalent resistance
across cap. terminals

Cut off freq

- freq at which gain becomes $\frac{1}{\sqrt{2}}$ times the max value or power becomes half of max value.
- also called as half power freq.

if max gain = A_v
 $(\text{gain})_{dB} = 20 \log A_v$

at cut off freq
gain = $A_v / \sqrt{2}$

$$\begin{aligned} \text{gain (dB)} &= 20 \log A_v / \sqrt{2} \\ &= 20 \log A_v - 20 \log \sqrt{2} \\ &= 20 \log A_v - 10 \log 2 \\ &= 20 \log A_v - 3 \text{ dB} \end{aligned}$$

3 dB frequency

cut off freq as
gain in
by 3 dB
max^m value



Cut off freq

- freq at which gain becomes $\frac{1}{\sqrt{2}}$ times the max value
or becomes half of max

as half power

if max gain = A_v
 $(\text{gain})_{dB} = 20 \log A_v$

at cut off freq

$$\text{gain} = A_v / \sqrt{2}$$

$$\begin{aligned} \text{gain (dB)} &= 20 \log A_v / \sqrt{2} \\ &= 20 \log A_v - 20 \log \sqrt{2} \\ &= 20 \log A_v - 10 \log 2 \\ &= 20 \log A_v - 3 \text{ dB} \end{aligned}$$

3 dB frequency

cut off freq as gain in dB reduces by 3 dB as comp. to max^m value.

$$\text{gain} = \frac{j\omega CR}{1 + j\omega CR}$$

$$|\text{gain}| = \frac{\omega CR}{\sqrt{1 + (\omega CR)^2}}$$

$$= \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}}$$

$$= \frac{1}{\sqrt{1 + \left(\frac{\omega_L}{\omega}\right)^2}}$$

RC

$$|\text{gain}| = \frac{1}{\sqrt{1 + \left(f_L/f\right)^2}}$$

f_L : lower cut off freq

$$f_L = \frac{1}{2\pi RC}$$

$$|\text{gain}| (f = f_L) = \frac{1}{\sqrt{2}} = \frac{\text{max. gain}}{\sqrt{2}}$$

$$|\text{gain}| (f \rightarrow \infty) = 1$$

$$\text{gain} = \frac{j\omega CR}{1 + j\omega CR}$$

$$\begin{aligned} |\text{gain}| &= \frac{\omega CR}{\sqrt{1 + (\omega CR)^2}} \\ &= \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}} \\ &= \frac{1}{\sqrt{1 + \left(\frac{\omega_L}{\omega}\right)^2}} \end{aligned}$$

$$\omega_L = 1/RC$$

$$|\text{gain}| = \frac{1}{\sqrt{1 + \left(f_L/f\right)^2}}$$

f_L : lower cut off

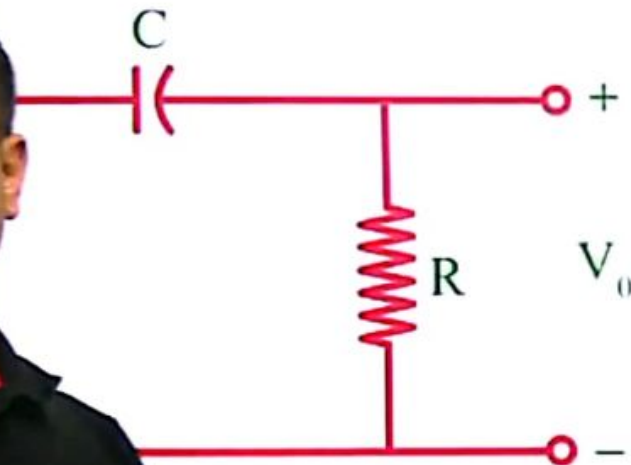
$$f_L = \frac{1}{2\pi RC}$$

12:05



Low frequency response

LF response of a BJT is similar to a high pass filter



$$X_C = \frac{1}{\omega C} = \text{high (open)}$$

$$\omega = 0$$

$$X_C \rightarrow 0 \text{ (short)}$$

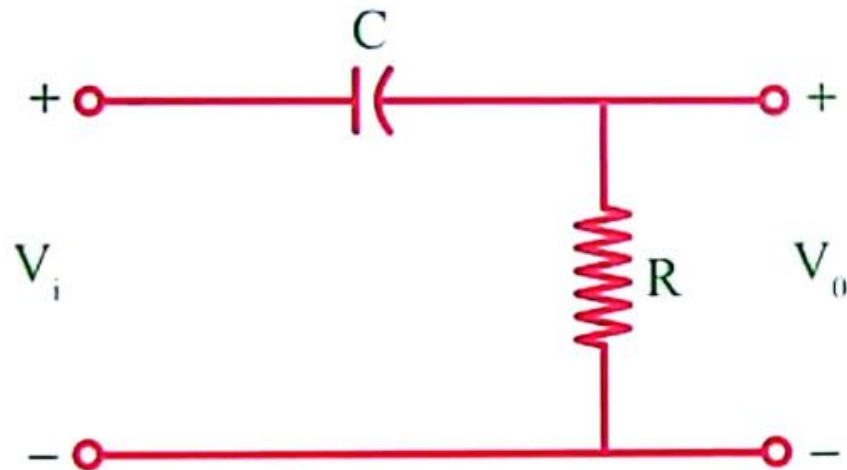
$$\begin{aligned} \frac{V_o}{V_i} &= \frac{R}{R + \frac{1}{sC}} \\ &= \frac{sCR}{1 + sCR} \end{aligned}$$

$$\frac{V_o}{V_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR}$$



Low frequency response

LF response of a BJT is similar to a high pass filter



at low freq $X_C = 1/\omega C = \text{high (open)}$

$$i = 0 \quad V_o = 0$$

at high freq $X_C \rightarrow 0$ (short)
 $V_o = V_i$

$$\frac{V_o}{V_i} = \frac{R}{R + X_C}$$

$$= \frac{\omega C R}{1 + \omega^2 C^2 R^2}$$

$$\frac{V_o}{V_i} \approx \omega C R$$

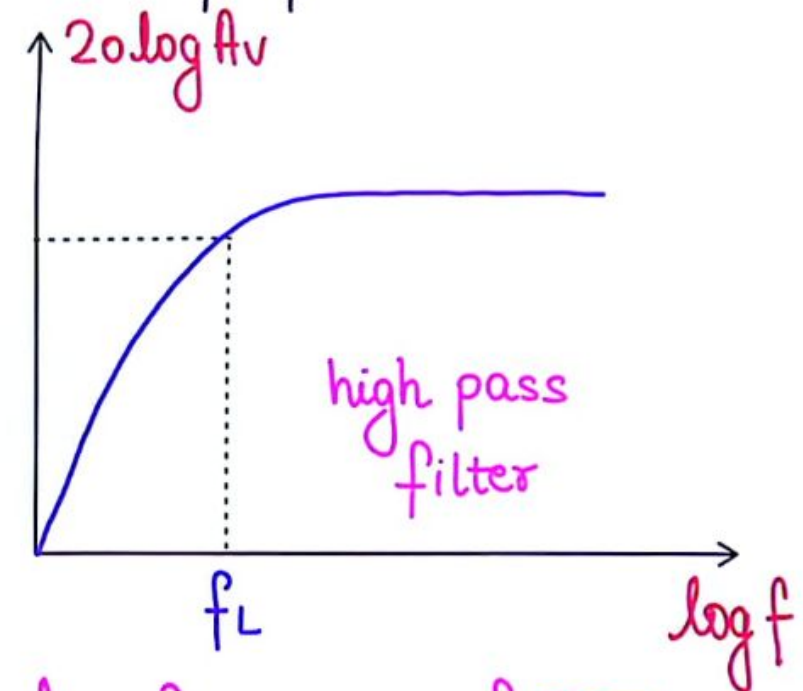
Q Why does gain of amplifier reduce at low freq?

→ at low freq, we consider C_{c1} , C_{c2} , C_E

→ due to C_{c1} & C_{c2} in series there is voltage drop so $V_b \downarrow$, gain $\downarrow$

→ C_E does not completely by-pass R_E $\therefore$ -ve f/b remains

gain $\downarrow$



low f response of BJT resembles a HPF

Q Why does gain of amplifier reduce at low freq?

→ freq. we consider

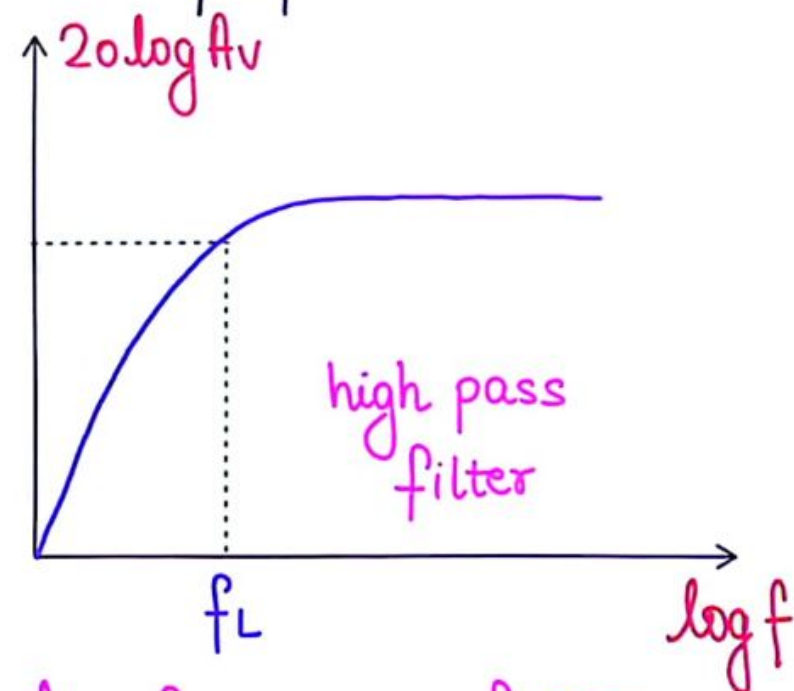
C_E

→ C_{c1} & C_{c2} in series

to voltage drop so

by-pass

as



low f response of BJT
resembles a HPF

Frequency Range	Coupling and Bypass Capacitor	Load and parasitic Capacitor
	C_{C1}, C_{C2}, C_E [μF] (low f) high	C_d, C_T, C_L [pF] (high f) low
Low	Consider $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{high}$ $X_C = \text{finite}$	open $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{low}$ $X_C = \text{high}$
	short $X_C = \text{low}$	open $X_C = \text{high}$
	short $\omega = \text{high}$ $C = \text{high}$ $X_C = \text{low}$	Consider $\omega = \text{high}$ $C = \text{low}$ $X_C = \text{finite}$